FlowRoute Logistics Analysis:

Business Insights and Recommendations

Project type: Simulation project (DataSimHQ)

Tools: PostgreSQL, Power BI, Excel

Data: 1.8M shipments, 8 hubs, Jan 2020 to Dec 2024

Conventions: money in naira (converted from kobo), "pp" means percentage points, every figure states its population and window.

1. Business problem:

Complaints were reported to have risen for three months, and the board wanted answers. Leadership could not tell whether the cause was a real operational failure, a specific hub, financial leakage, or errors in the data itself.

2. Executive Summary:

- No hub, complaint category or customer type is an **outlier** in any analysis. The problem is **systemic**, so the fixes are process and system changes, not action against particular sites or customer groups.
- The **~10%** failure rate is real. A driver-app defect inflated it by only **0.19 pp**.
- Every complaint (**64,737**) sits on a failed or returned shipment. About 30% of those shipments generate a complaint. Delivered and in-flight shipments produce none, so failed and returned deliveries are the whole complaint problem.
- The complaint "spike" is not supported by the data. The complaint rate has been about 36 per 1,000 shipments every year from 2020 to 2024 (35.5 to 36.3).
- Failure is uniform. It is about 10% at every hub, for every driver (spread exactly as chance predicts) and in every month.

3. The Real Business Problem:

The question the board asked was whether complaints were caused by a bad hub, financial leakage, data errors or real operational failure. The analysis shows the real problem is none of the first three. It is a **systemic weakness in how shipments are completed and recorded across the whole network**:

- Only **72.0%** of shipments reach delivered. About **10%** fail, and the rest are in flight, returned, created or cancelled.
- **All 64,737 complaints sit on failed (53,897) or returned (10,840) shipments**, and about 30% of those shipments produce a complaint. Complaints are split about 25% each across lost, late delivery, wrong address and damaged, at about 36 per 1,000 shipments at every hub.
- Internal reports do not reconcile with each other. Recorded COD includes amounts on failed, returned, cancelled and stuck parcels.
- These problems are **the same at every hub, for every customer type, for every driver and in every year**, so they come from process and system design, not from a specific site, group or person.
- **The complaint rise is not supported by the data.** Complaints per 1,000 shipments were 36.0, 36.1, 35.5, 35.8 and 36.3 in 2020 to 2024. October to December 2024 volume was flat (1,107, 1,090, 1,099).
- The accurate statement is: "The data shows a steady complaint rate of about 36 per 1,000 shipments, produced entirely by the roughly 12% of shipments that fail or are returned. It does not show a recent spike."

4. Hypotheses: Confirmed, & Rejected:

#	Hypothesis	Verdict	Evidence
1	A specific hub is underperforming.	Rejected	Delivery rate 71.90% to 72.08%; complaints 35.7 to 36.5 per 1,000; transit differs by 0.07 hrs; no hub stands out in any finding.
2	One complaint category dominates.	Rejected	Categories range 24.7% to 25.6%
3	Business accounts cause the weight leakage.	Rejected	Same under-declaration rate (20.66% vs 20.72%) and same ₦ per parcel (about ₦2,350) as individuals.
4	Missing delivery-address coordinates cause failed deliveries.	Rejected	Failure rate 10.00% with coordinates vs 9.95% without; transit 61.89 vs 61.92 hrs.
5	The reported failure rate is inflated by a software defect.	Confirmed but small	Defect on 3,447 shipments; rate 9.99% to 9.80%. The ~10% failure rate is real.
6	Timestamp errors distort	Confirmed but	12,335 delivered records with

	delivery time.	small	delivery before pickup; average moves 0.61 hrs.
7	The audit COD figure is overstated by data errors.	Confirmed	₦313.9M on 62,254 non-COD shipments (0.58%).
8	Financial leakage exists.	Confirmed	About ₦152M a year of weight under-declaration, steady since 2020.
9	Real operational failure is the root cause.	Supported	100% of complaints (64,737) sit on failed or returned shipments; about 30% of those shipments produce a complaint.
10	Complaints rose over the last three months.	Not supported	About 36 per 1,000 every year (35.5 to 36.3); Oct to Dec 2024 volume flat.
11	Stuck parcels drive the "lost" complaints.	Rejected	Zero complaints on in-transit, created or out-for-delivery shipments.
12	Specific drivers cause failures.	Rejected	4,200 drivers: spread of failure rates 1.45 pp, the same as the 1.45 pp expected by chance; 4 drivers more than 3 spreads above average vs 5.7 expected.
13	Complaint categories are reliable.	Doubtful	Each category is about 25% on failed shipments and about 25% on returned ones.

Overall: hypotheses 1 to 4 and 11 to 12 are rejected, 5 to 8 are confirmed but do not explain the complaints, 9 is supported, 10 is not supported. Hypothesis 13 is flagged because category labels look unreliable.

5. Business Insights:

Insight 1: Complaints come only from failed and returned shipments.

- 64,737 complaints, all on failed (53,897) or returned (10,840) shipments. About 30% of failed shipments (30.0%) and returned shipments (30.2%) produce a complaint. Delivered, in-transit, created, out-for-delivery and cancelled shipments (1.58M) produce none.
- The complaint rate is steady: 36.0, 36.1, 35.5, 35.8 and 36.3 per 1,000 shipments in 2020 to 2024.
- Hub complaint rates range 35.7 to 36.5 per 1,000 shipments. Lagos vs Kano: 36.1 vs 35.8, effectively the same.

- Categories split almost equally: lost 25.6%, wrong address 24.9%, late delivery 24.8%, damaged 24.7%.
- 65.1% of complaints are resolved, so about 35% remain unresolved.
- **Implication:** fixing the complaint problem means reducing failed and returned deliveries and responding to them. The app defect, weight leakage, COD errors and stuck parcels do not generate complaints.

Insight 2: The failure rate is real; the app defect is small.

- Raw 9.99% (179,859 failed) vs corrected 9.80% (176,412).
- A driver-app defect let delivery attempts exceed the policy maximum on **3,447 shipments**, all with status failed. It ran from Jan 2020 to **14 Mar 2023** (no cases afterwards), at every hub (11.5% to 13.1% each).
- The correction assumes those shipments were not genuinely failed. That is untested, so the defensible statement is a **range of 9.80% to 9.99%**.
- Monthly failure rate is flat (9.4% to 10.3%), and 2024 shows 10.08%.
- **Implication:** the board should not act on the idea that failures are a software artefact. About 176,000 failed shipments remain after correction.

Insight 3: Weight under-declaration costs roughly ₦152M a year, and business accounts are not the cause.

• 20.7% of weighed shipments exceed their declared weight by more than 20% (323,752 shipments, ₦760.9M all-time). Including smaller excesses, the figure computed in the board pack is ₦1,017.4M.

- **Last 12 months:** 64,673 shipments, ₦152.25M (>20% tier).
- Business accounts under-declare at the same rate as individuals (20.66% vs 20.72%) and with the same average shortfall (about ₦2,350 per parcel).
- The rate is stable (20.71% all-time vs 20.65% in the last 12 months).
- **Judgement call:** the >20% threshold was chosen to filter scale and packaging noise.
- **Implication:** the pattern points to a process gap, not to specific customers or sites.

Insight 4: Recorded COD is overstated by ₦313.9M because non-COD shipments carry a COD amount.

- Raw ₦54.61bn vs corrected ₦54.30bn, an overstatement of ₦313.9M (0.58% of the corrected figure).

- 62,254 shipments with payment method prepaid (40,034), credit (13,397) or wallet (8,823) carry a COD amount averaging about ₦5,000, against about ₦100,600 for genuine COD.
- About 1,000 shipments a month since Jan 2020, **still occurring**, and identical at every hub (3.40% to 3.51% of shipments).
- **Implication:** the audit report sums COD amounts across every shipment, so a validation rule at data entry would remove the error.

Insight 5: The audit report's "exposure" is far larger than the cash actually at risk.

Even after removing phantom COD, the ₦54.30bn is a five-year cumulative sum of recorded amounts, not an exposure:

Component	Amount	Comment
Failed, returned, cancelled	₦7.05bn	No cash was collected on these.
In-flight over 90 days	₦7.75bn	256,625 shipments (95% of in-flight); 34,381 are still at created. These are stuck, not in transit.
In-flight 0 to 90 days	about ₦0.41bn	The genuinely live amount.
Delivered	₦39.09bn	Only ₦2.51bn has a completed cod cash.

Insight 6: A defensible delivery time is 61.9 hours, and hubs do not differ.

- Average **61.90 hrs** (median 61.86) across 1,283,555 delivered shipments, **99.05%** of all delivered shipments. Raw average (including impossible records) is 61.29 hrs.
- 12,335 delivered shipments have delivery before pickup. Across all statuses there are 14,372 (the other 2,037 are failed or returned shipments carrying a delivery timestamp, a separate defect).
- The records run from the start of the data to **2 Jun 2022**, matching the board report of an app fix in June 2022.
- **Missing coordinates:** 55,974 delivery addresses (360,509 shipments, 20%). Failure rate is 10.00% with coordinates and 9.95% without, and transit is 61.89 vs 61.92 hrs.

Insight 7: Failure is uniform across hubs, drivers and months.

- About 10% of shipments fail at every hub (71.90% to 72.08% delivered), in every month (9.4% to 10.3%) and in every year.
- **Drivers:** 4,200 drivers with about 428 shipments each. The observed spread of driver failure rates (1.45 pp) equals the spread expected from chance alone (1.45 pp), and 4 drivers sit more than 3 spreads above average against 5.7 expected by chance. The top-ranked driver (15.6%) is luck, so **ranking or blaming individual drivers on this data would misidentify them**.
- Missing delivery-address coordinates (20% of shipments) have no effect on failure rate.
- **Implication:** failure behaves like a uniform baseline built into the network.

6. Recommendations:

Recommendation	Evidence	KPI to track
Investigate and close out stuck shipments; add a status timeout so no parcel stays in flight for months.	256,625 shipments over 90 days; ₦7.75bn in COD amounts. These produce no complaints, so this is a reporting and cash risk.	Shipments in-flight over 30 days.
Re-bill weight differences automatically when the hub weighs a parcel.	About ₦152M a year on parcels over 20% above declared weight, at a stable rate.	Weight leakage per 1,000 shipments; % re-billed.
Block a COD amount on non-COD shipments at data entry.	~1,000 shipments a month since Jan 2020.	Phantom COD shipments per month (target: 0).
Fix timestamp capture (one timezone, validation that delivery follows pickup) and write the rule in the data dictionary.	12,335 impossible delivered records; 2,037 failed or returned shipments with a delivery timestamp.	Records with delivery before pickup.
Use rates, not counts, in board reporting, with the window and population on every figure.	Total vs per-1,000 comparisons changed several conclusions	n/a
Do not take action against individual hubs, drivers or customer groups on current evidence.	No hub, customer or driver outlier; driver variation equals chance	n/a
Reduce failed and returned deliveries, and contact customers proactively when a	100% of complaints sit on failed or returned shipments; about 30% of them produce	Failed plus returned rate; complaints per 1,000 failed or

delivery fails (re-delivery offer, status message)	a complaint.	returned shipments.
Investigate why deliveries fail (shipment characteristics, routes, tracking exceptions) before committing to a fix.	Failure is uniform at about 10%, and the cause is not explained by hub, customer type, coordinates or driver.	Failure rate by cause category.

7. Raw vs Corrected Figures:

Finding	Raw	Corrected	Difference	Reason
Failure rate	9.99%	9.80%	0.19 pp	Attempts above policy on 3,447 failed shipments.
Weight leakage (all-time)	₦1,017.4M	₦760.9M (>20% over)	₦256.5M	Threshold excludes excesses of 20% or less.
Recorded COD	₦54.61bn	₦54.30bn	₦313.9M	COD amount on 62,254 non-COD shipments.
Avg delivery time	61.29 hrs	61.90 hrs	0.61 hrs	12,335 delivered shipments with delivery before. pickup

8. Why the Excel results differ from SQL and Power BI?

SQL and Power BI ran on the **full dataset** (1.8M shipments). Excel could not hold that volume, so the Excel files use **extracts: a sample or a shorter time window**.

Episode	Excel result	SQL / Power BI result	Why they differ
Complaints	Same counts (64,737; category and hub tables)	Same	Full data in both; match.
Failure rate	3,447 over cap; 9.99% to 9.80%	Same	Full data in both; match exactly.
Weight leakage	12 months: ₦152,250,395; 2 years: ₦304.1M	All-time ₦760.9M; 12 months ₦152.25M	Different time windows. The 12-month figure matches SQL and Power BI.
COD	₦10.95bn to ₦10.88bn; phantom ₦63.3M (12,511 shipments)	₦54.61bn to ₦54.30bn; phantom ₦313.9M (62,254 shipments)	Different time windows (12 months vs all-time). The overstatement is 0.58% in both.

Transit time	Extract of 26,372 rows: 27.2 to 62.3 hrs ("35-hour distortion")	61.29 to 61.90 hrs (0.61-hr difference)	Sample, not population. The extract held all 14,372 impossible records but only 12,000 valid ones, so it exaggerated the bug about 57 times
Excel formula	AVERAGEIFS gives about 62.09 hrs	61.90 hrs	The table holds a sample of about 10,000 delivered shipments. A gap of 0.2 hrs is within normal sampling error (rough estimate)
Board summary tab	Mixed windows and populations	Consistent full-data figures	The tab must be replaced with the SQL and Power BI figures.

9. Analytical decisions:

- Compared rates, not totals, so larger groups were not blamed for their volume.
- Cross-checked every headline figure in SQL, Excel and Power BI. Differences traced to time windows, timezone boundaries and an Excel sample that held 14,372 bad rows but only 12,000 good rows, which exaggerated the transit-time bug about 57 times.
- Treated the three data-error corrections differently from the weight threshold, which is a judgement call.
- Restricted the delivery-time metric to delivered shipments, after finding the first version also included failed and returned shipments.
- Tested drivers by comparing observed variation with the variation expected from chance, instead of ranking drivers on small samples.
- Linked complaints to shipment status, which showed that all complaints sit on failed or returned shipments and corrected my earlier assumption that stuck parcels drive complaints.